

Department of Electronics and Electrical Engineering
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EE101 - Basic Electronics

Tutorial 2

Date: Aug 19, 2023

1. Refer to the electric circuit shown in Figure 1. Perform *Nodal Analysis* and express the node voltages (w.r.t. reference node R) in the compact $V = [v_a \ v_b \ v_c \ v_d]^T = Y^{-1}CU$ form (usual notations). Use the following numerical values and evaluate the matrix Y and vector CU . $R_1 = 20\Omega$, $R_2 = 40\Omega$,

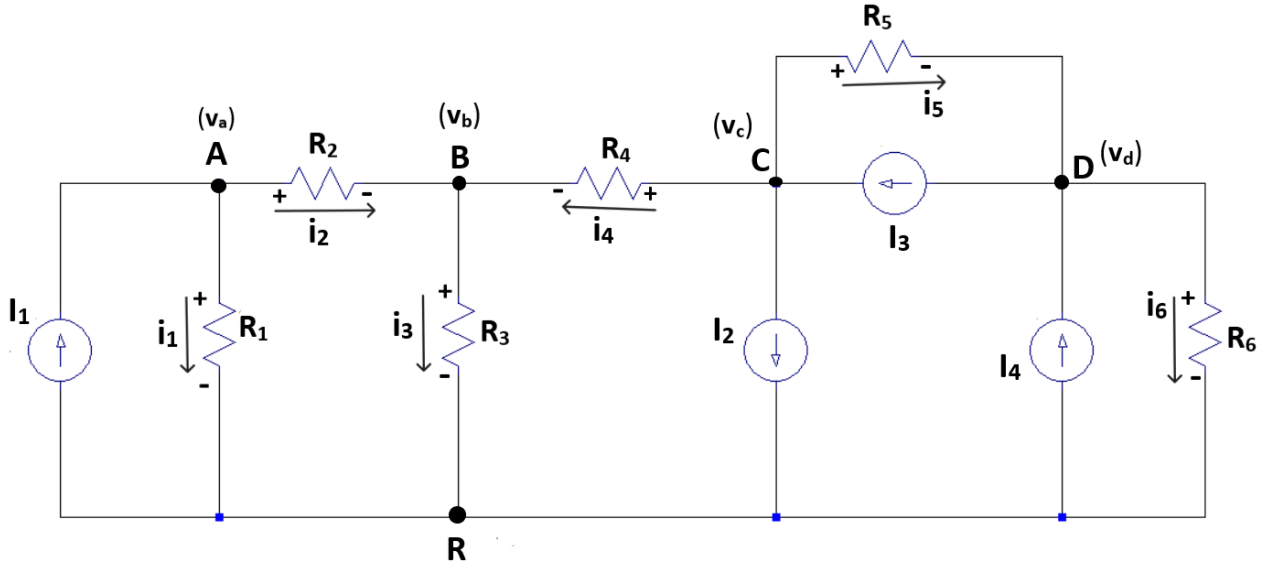


Figure 1: Electric circuit with independent current sources.

$$R_3 = 100\Omega, R_4 = 50\Omega, R_5 = 10\Omega, R_6 = 200\Omega, I_1 = 10A, I_2 = 2.5A, I_3 = 2A, I_4 = 5A.$$

2. Refer to the electric circuit shown in Figure 2. The node voltages V_a , V_b , V_c and V_d are w.r.t node R . Construct the equation $YV = CU$ in terms of the circuit element values.

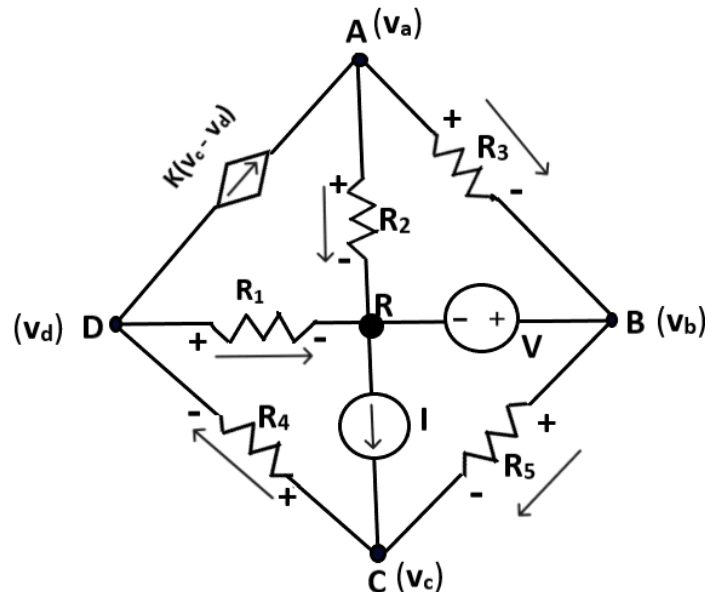


Figure 2: Electric circuit with both independent and dependent sources.

3. The majority function $M(X, Y, Z)$ is equal to 1 only when two or three of its arguments equal 1. Consequently, the function is equal to 0 when two or three of its arguments equal 0.
- Express the function $M(X, Y, Z)$ as a sum of minterms (Standard SOP).
 - Express the function $M(X, Y, Z)$ as a product of maxterms (Standard POS).
 - Show that these expressions can be simplified as

$$M(X, Y, Z) = X.Y + X.Z + Y.Z = (X + Y)(X + Z)(Y + Z)$$

- Argue that $M(A, B, M(C, D, E)) = M(M(A, B, C), D, M(A, B, E))$.
- For the function $F(A, B, C, D)$ corresponding to the network of Figure 3, find algebraic expressions in the SSOP and SPOS form.

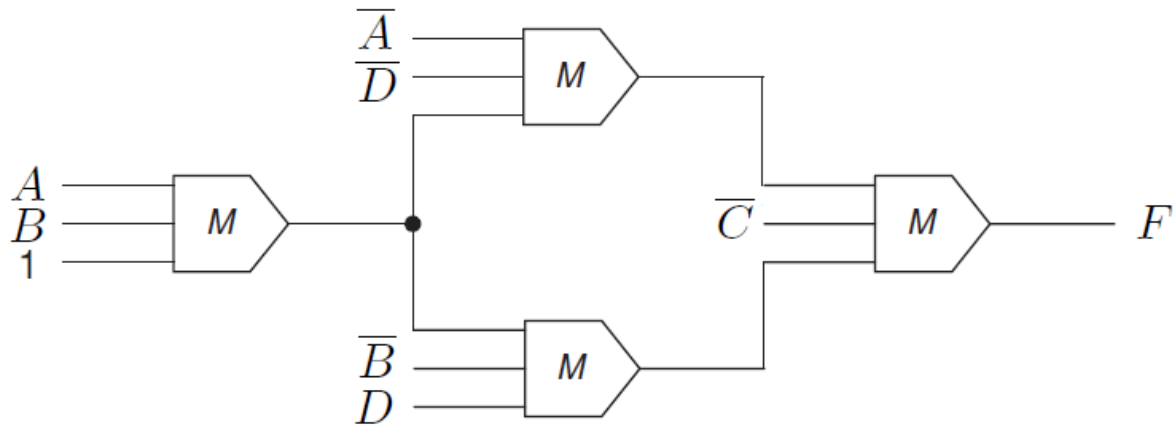


Figure 3: Digital circuit built using the majority function

- Simplify the expressions obtained in the previous step.
4. Simplify the following Boolean expressions as much as possible using basic Boolean identities:
- $S(A, B, C) = \overline{A}.\overline{B}.C + B.C + A.C$
 - $T(A, B, C, D) = \overline{A}.B + A.B.D + A.\overline{B}.C.\overline{D} + B.C$
 - $U(A, B, C, D, \dots) = A + \overline{A}.B + \overline{A}.\overline{B}.C + \overline{A}.\overline{B}.\overline{C}.D + \dots$
 - $V(A, B, C) = A.B.C + \overline{A}.\overline{B}.C + \overline{A}.B.\overline{C} + A.\overline{B}.\overline{C}$